CCCA Meta-Atlas — Raghavan et al. 2021 (Pancreatic Ductal Adenocarcinoma, Seq-Well)

GO / NO-GO GATES

Gate 1 · mixing & continuity — **GO**

Harmony θ-sweep resolves a per-patient archipelago (θ=0, iLISI≈1) into one continuous well-mixed mass (θ=3, iLISI 4.8/21). Island-free (largest component 99.1%; 72-cell speck removed).

Gate 2 · module geography — **GO (non-branching / linear)**

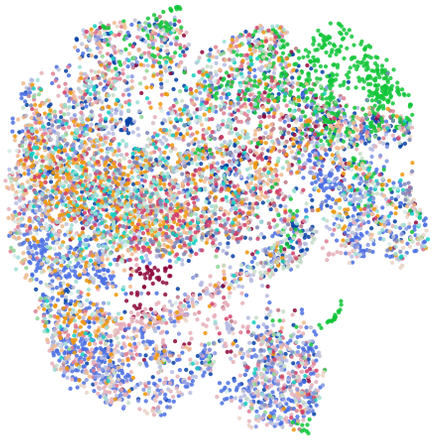
Single embryonic axis: Gast/pSom co-vary at the origin (r=+0.12), both oppose Crest toward the terminus (Gast–Crest −0.50, pSom–Crest −0.61; pSom&Crest joint-high enrichment 0.16). No two-arm hub.

Gate 3 · hub cross-repression — **NA (no bifurcation)**

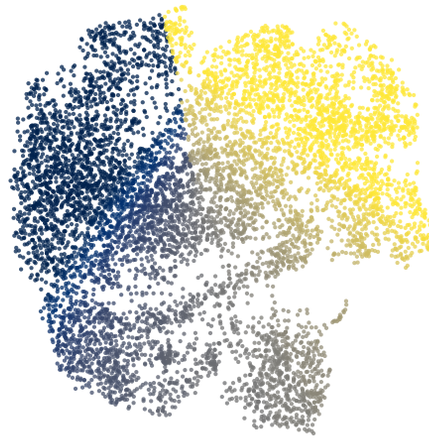
Curve has 0 forks (PPT tree forks unstable 2→3 across N — round-blob overfit). No branch to test.

STEP FIGURES

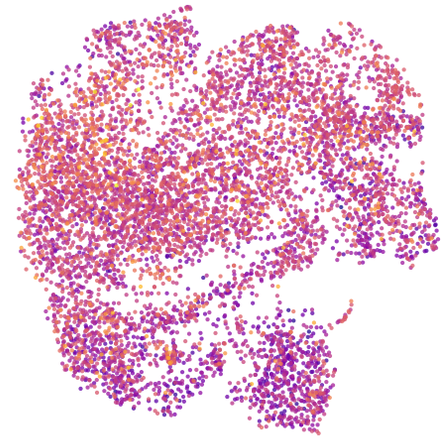
Gate 1: sample integration



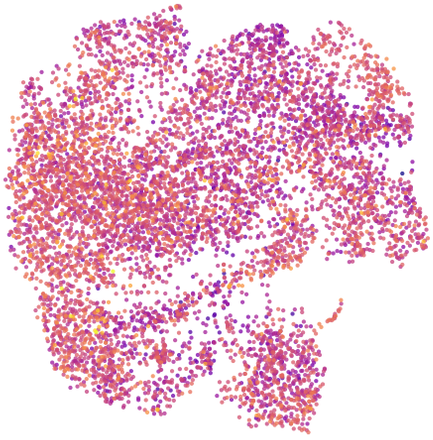
Trajectory (root → terminus)



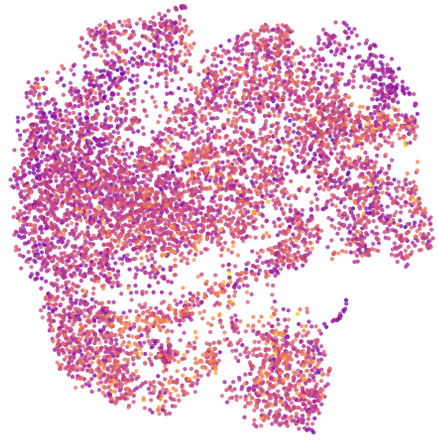
Gast signature



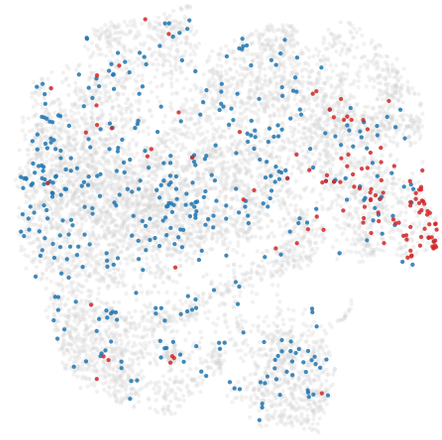
pSom signature



Crest signature



EMT-TFs: FOXC1 (early) · ZEB2 (late)



Dataset. Metastatic PDAC (Raghavan et al., 2021), Seq-Well. 7,904 malignant cells → 7,830 after ≥100/sample threshold (21/23 samples) → 7,758 after island removal. Subtypes: Classical 3,314 · Mixed 3,341 · Basal 1,103.

Verdict. The embryonic pattern reproduces as the non-branching case — one continuous Gast/pSom-origin → Crest-terminus axis with pSom/Crest mutual exclusivity; developmental EMT-TF order FOXC1 → ZEB2. No bifurcation.

PARAMETER CHOICES

Min cells / sample	≥100 (gap 45→120; 21/23 kept)	**Trajectory**	scFates curve — Nodes=20, λ=0.3, μ=100
Integration	Harmony θ=3 · n_pcs=30 · spread=1.3	**Root**	tip 0 = Gast/pSom-high, Crest-low (dev. origin)
Gene selection	1,497 overdispersed (find_overdispersed)	**Fold check**	self-approach 2.56 (>1 fold-free); 0 forks
Scope	whole manifold (cycling |ρ|≤0.19, orthog.)	**EMT-TF panel**	Snail/Zeb/Twist + FOXC/PRRX, gated >3% expr
Subtype source	Raghavan 2021 · Moffitt basal/classical	**Module means**	Gast −0.011 · pSom +0.072 · Crest −0.024

Env: scFates conda env (R_HOME set in-notebook). Notebooks 01–04 run in place and reproduce every value above. Attempt 4 ("-3").